

REAYER RPO Simulation

Equation Reference

Summary of all equations used in `rpo\_dynamics.py`,
`rpo\_guidance.py`, and `rpo\_control.py`.

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1. Orbital Mechanics — Mean Motion

All CW relative-motion equations are parameterised by the mean motion n of the reference (target) orbit.
At GEO ($a = 42\,878\text{ km}$) n is approximately $7.11 \times 10^{-5}\text{ rad/s}$, roughly 15× smaller than a typical LEO value.

Mean motion of a circular orbit

$$n = \sqrt{\frac{\mu}{a^3}}$$

$$\mu = 3.986 \times 10^{14}\text{ m}^3/\text{s}^2 \text{ (Earth gravitational parameter)}$$

$$a = \text{semi-major axis of the target orbit [m]}$$

REAPER values: $a = 42\,878\text{ km}$, $n = 7.11 \times 10^{-5}\text{ rad/s}$
Orbital period $T = 2\pi/n \approx 88\,341\text{ s} \approx 24.5\text{ h}$

2. Relative Translational Dynamics (Hill / CW Equations)

The Clohessy-Wiltshire (Hill) equations describe the relative motion of the chaser with respect to the target in the local Hill frame:

x = radial (R -bar), y = along-track (V -bar), z = cross-track (H -bar).

Assumptions: target on a circular orbit; inter-satellite distance $\ll$ orbital radius.

Hill / CW equations of relative motion

$$\ddot{x} = \ddot{x} - 2n\dot{y} - 3n^2x = f_x$$

$$\ddot{y} = \ddot{y} + 2n\dot{x} = f_y$$

$$\ddot{z} = \ddot{z} + n^2z = f_z$$

x, y, z = relative position components [m]

f_x, f_y, f_z = specific force (thrust + perturbations) [m/s²]

n = mean motion of target orbit [rad/s]

State vector form used in the simulation: $\mathbf{x} = (x, y, z, \dot{x}, \dot{y}, \dot{z})^T$

Perturbation forces at GEO (replaces LEO drag/magnetic from paper)

Atmospheric drag: negligible at GEO altitude (42 878 km). Magnetic dipole torque: negligible. Two forces remain:

$$a_{SRP} = \frac{C_R \cdot (A/m) \cdot P_{\odot}}{1}$$

a_{triax} from Earth J_{22} tesseral harmonic (GEO longitude-dependent); representative value $\approx 3 \times 10^{-8} \text{ m/s}^2$

C_R = 1.5 (reflectivity, MLI-covered bus)

A/m = 0.015 m²/kg (MS area-to-mass ratio, placeholder)

$P_{\odot}$ = $4.56 \times 10^{-6} \text{ N/m}^2$ (solar radiation pressure at 1 AU)

3. CW State-Transition Matrix (STM)

Under zero forcing (free drift), $\mathbf{x}(t) = \Phi(t) \mathbf{x}(0)$ where $\Phi(t)$ is the 6×6 STM.

Shorthand: $s = \sin(nt)$, $c = \cos(nt)$, $\mathbf{x} = (x, y, z, \dot{x}, \dot{y}, \dot{z})^T$

Position-to-position block Φ_{rr} (3x3)

$$\begin{aligned}\Phi_{rr}[1, 1] &= 4 - 3c \\ \Phi_{rr}[2, 1] &= 6(s - nt) \\ \Phi_{rr}[3, 3] &= c \\ \text{all other entries} &= 0\end{aligned}$$

Position-to-velocity block Φ_{rv} (3x3)

$$\begin{aligned}\Phi_{rv}[1, 1] &= s/n \\ \Phi_{rv}[1, 2] &= 2(1 - c)/n \\ \Phi_{rv}[2, 1] &= 2(c - 1)/n \\ \Phi_{rv}[2, 2] &= (4s - 3nt)/n \\ \Phi_{rv}[3, 3] &= s/n, \text{ all other entries} = 0\end{aligned}$$

Velocity-to-position block Φ_{vr} (3x3)

$$\begin{aligned}\Phi_{vr}[1, 1] &= 3ns \\ \Phi_{vr}[2, 1] &= 6n(c - 1) \\ \Phi_{vr}[3, 3] &= -ns, \text{ all other entries} = 0\end{aligned}$$

Velocity-to-velocity block Φ_{vv} : $\Phi_{vv}[1, 1] = c$, $\Phi_{vv}[1, 2] = 2s$, $\Phi_{vv}[2, 1] = -2s$,
 $\Phi_{vv}[2, 2] = 4c - 3$, $\Phi_{vv}[3, 3] = c$, all other entries = 0

Implementation: `rpo_dynamics.cw_stm(n, t) → full 6x6 matrix.`

Used for: free-drift propagation, two-impulse targeting, path sampling for plots.

4. Two-Impulse CW Targeting & Transfer Time

For each translational phase, find the two burns that transfer the chaser from rest at $\mathbf{r}_0$ to rest at $\mathbf{r}_f$ in time t .

Two-impulse CW targeting

Step 1: $\mathbf{v}_0^+ = \Phi_{rv}^{-1}(\mathbf{r}_f - \Phi_{rr} \mathbf{r}_0)$

Solve for the required velocity just after the first burn. Φ_{rv} must be invertible (non-singular transfer).

Step 2: $\Delta \mathbf{v}_1 = \mathbf{v}_0^+ - \mathbf{v}_0$

First burn: difference between required and current velocity.

Step 3: $\mathbf{v}_f^- = \Phi_{vr} \mathbf{r}_0 + \Phi_{vv} \mathbf{v}_0^+$

Velocity at arrival, just before the second burn.

Step 4: $\Delta \mathbf{v}_2 = \mathbf{v}_f - \mathbf{v}_f^-$

$$\Delta V = |\Delta \mathbf{v}_1| + |\Delta \mathbf{v}_2|$$

Transfer time sizing

chord = $d = |\mathbf{r}_f - \mathbf{r}_0|$

t = $t = \frac{d}{0.5 V_{max}}$

The transfer time is sized so the mean speed equals $0.5 V_{max}$, ensuring the velocity constraint $|\mathbf{v}(t)| \leq V_{max} = 1.0 \text{ m/s}$ (low-impact docking requirement) is met with margin.

$$V_{max} = 1.0 \text{ m/s} \quad (\text{maximum approach velocity})$$

$$t_{min} = 1.0 \text{ s} \quad (\text{hard lower bound to avoid STM singularity})$$

5. GEO Perturbations & Stationkeeping ΔV

During hold / dwell phases (P2, RH A1) the chaser must null the accumulated drift from two persistent GEO perturbations. No net translation occurs.

Solar Radiation Pressure (SRP)

$$a_{SRP} = C_R \cdot \frac{A}{m} \cdot P_{\odot}$$

$$C_R = 1.5 \quad \text{reflectivity coefficient (MLI)}$$

$$A/m = 0.015 \text{ m}^2/\text{kg} \quad \text{area-to-mass ratio (placeholder)}$$

$$P_{\odot} = 4.56 \times 10^{-6} \text{ N/m}^2 \quad \text{solar pressure at 1 AU}$$

$$\text{Result: } a_{SRP} \approx 1.03 \times 10^{-8} \text{ m/s}^2$$

Earth Triaxiality (J₂₂ tesseral harmonic)

A longitude-dependent tangential acceleration from Earth's equatorial ellipticity. Represented as a constant bias per phase.

$$a_{triax} \approx 3.0 \times 10^{-8} \text{ m/s}^2$$

Stationkeeping ΔV (dwell phases)

First-order model: constant disturbance integrates into drift velocity that must be cancelled once per dwell.

$$a_{dist} = a_{SRP} + a_{triax}$$

$$\Delta V_{SK} = a_{dist} \cdot t_{dwell}$$

$$t_{dwell} = 28\,800 \text{ s (8 h, confirmed)} \rightarrow \Delta V_{SK} \approx 0.004 \text{ m/s (negligible for } \Delta V \text{ budget, \n significant for } T_{OPS} \text{ update)}$$

6. Rigid-Body Rotational Dynamics (Euler's Equation)

*Used to propagate the tumbling state of the target debris.
The chaser's own attitude is controlled, not freely tumbling.*

Euler's rigid-body equation

$$\dot{\boldsymbol{\omega}} = \boldsymbol{\omega} = \mathbf{J}^{-1}(\mathbf{M}_{\text{net}} - \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega})$$

$\boldsymbol{\omega}$ = angular velocity vector of the body [rad/s]

$\mathbf{J}$ = instantaneous inertia tensor [kg·m²]

$\mathbf{M}_{\text{net}}$ = net external torque vector [N·m]

$\mathbf{M}_{\text{net}} = \mathbf{0}$ for free tumble (target debris)
The target is modelled as free-tumbling ($\mathbf{M}_{\text{net}} = \mathbf{0}$) at the requirement ceiling of 1 rpm (6 deg/s) about an arbitrary axis.

*Integration: `scipy.integrate.solve_ivp` with RK45,
`rtol = 10-9`, `atol = 10-12`, quaternion renormalised every step.*

Conservation check — rotational kinetic energy

*Under free tumble with no external torques, rotational KE is conserved.
Used as a numerical verification of the integrator.*

$$T_{\text{rot}} = T_{\text{rot}} = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{J} \boldsymbol{\omega} = \text{const}$$

Test passes if $\sigma(T_{\text{rot}})/\mu(T_{\text{rot}}) < 10^{-3}$ over the propagation interval.

7. Quaternion Kinematics

Attitude is represented by the unit quaternion $\mathbf{q} = (q_1, q_2, q_3, q_4)^T$ where q_4 is the scalar part. Propagated alongside the Euler equation.

Quaternion kinematic equation (Conings & Mooij Eq. 3)

$$\dot{\mathbf{q}} = \frac{1}{2} \Xi(\mathbf{q}) \boldsymbol{\omega}$$

$\Xi(\mathbf{q})$ — the 4×3 kinematic matrix

Row 1: $[q_4, -q_3, q_2]$
 Row 2: $[q_3, q_4, -q_1]$
 Row 3: $[-q_2, q_1, q_4]$
 Row 4: $[-q_1, -q_2, -q_3]$

$\Xi(\mathbf{q})$ is 4×3. The product $\Xi \boldsymbol{\omega}$ yields a 4-vector $\dot{\mathbf{q}}$.

After each integration step the quaternion is renormalised:
 $\mathbf{q} \leftarrow \mathbf{q}/|\mathbf{q}|$ to prevent numerical drift from the unit-norm constraint.

Combined state vector for target tumble propagation

State: $\mathbf{y} = (\mathbf{q}, \boldsymbol{\omega})^T \in \mathbb{R}^7$

$$\dot{q}_1, \dot{q}_2, \dot{q}_3, \dot{q}_4 = \frac{1}{2} \Xi(\mathbf{q}) \boldsymbol{\omega}$$

$$\dot{\boldsymbol{\omega}} = \mathbf{J}^{-1}(\mathbf{M}_n \mathbf{e}_t - \boldsymbol{\omega} \times \mathbf{J} \boldsymbol{\omega})$$

8. Body Geometry & Inertia Tensors

Uniform rectangular box (used for MS, tug, debris bus)

$$I_{xx} = I_{xx} = \frac{m}{12}(b^2 + c^2), \quad I_{yy} = \frac{m}{12}(a^2 + c^2), \quad I_{zz} = \frac{m}{12}(a^2 + b^2)$$

a, b, c = edge lengths along x, y, z [m]

MS: $3.0 \times 3.0 \times 3.5$ m, Tug: $0.8 \times 0.8 \times 0.8$ m (both confirmed)

Debris bus: $2.5 \times 2.5 \times 3.0$ m (placeholder)

Debris inertia with deployed solar panels

Panels modelled as point masses at $\pm$ half-span along the body y -axis. They inflate I_{xx} and I_{zz} while leaving I_{yy} near the bus value:

$$J_{debris} = J_{bus} + m_{panel} \cdot \left(\frac{s}{2}\right)^2 \cdot \text{diag}(1, 0, 1)$$

$m_{panel} = 0.15 \times m_{debris}$ (both panels combined, 15% mass fraction)

$s = 17.2$ m (COMSATBW-1, BSS-702, confirmed solar panel span)

Parallel-axis theorem (combined body assembly)

To shift a body's inertia from its own CoM to a reference point:

$$J_{ref} = J_c m + m(|\mathbf{d}|^2 \mathbf{I} - \mathbf{d} \otimes \mathbf{d})$$

$\mathbf{d}$ = vector from reference point to body CoM. Applied to every sub-body to build the combined (MS + arm + tug + debris) inertia for the detumble phase.

9. Attitude Control Budget (Angular Impulse Method)

*Reaction wheels carry smooth tracking torque but cannot hold net spin indefinitely.
The net angular momentum change is therefore delivered (or dumped) by the RCS thrusters.
This gives a conservative (all-thruster) propellant bound — correct posture for sizing.*

Angular momentum and torque

$$H = H = |\mathbf{J} \boldsymbol{\omega}|$$

$$\tau = \tau = 2FL$$

Opposed thruster pair: two thrusters each at force F , separated by moment arm $L \rightarrow$ torque $\tau = 2FL$.

$$F = 64 \text{ N (RCS thruster force, confirmed)}$$

$$L = 1.75 \text{ m (effective moment arm, confirmed)}$$

Propellant and equivalent ΔV

$$m_{prop} = m_{prop} = \frac{H}{L \cdot v_{ex}}$$

$$\Delta V_{att} = \Delta V_{att} = v_{ex} \ln \left(\frac{m_0}{m_0 - m_{prop}} \right)$$

$$t_{burn} = t_{burn} = \frac{H}{2FL}$$

Phase 4 (arm extension) uses the momentum delta between two configurations:

$$\Delta H = \Delta H = |H_{ready} - H_{stowed}|$$

$$H_{stowed} = |\mathbf{J}_M \mathbf{S} \boldsymbol{\omega}| \text{ (tug not yet extended)}$$

$$H_{ready} = |\mathbf{J}_M \mathbf{S} + \text{tug} \boldsymbol{\omega}| \text{ (tug on arm at full extension, debris NOT coupled)}$$

10. Propellant Budget & Tsiolkovsky Equation

Tsiolkovsky rocket equation

$$\Delta V = v_{ex} \ln \frac{m_0}{m_0 - m_{prop}}$$

$$m_{prop} = m_0(1 - e^{-\Delta V/v_{ex}})$$

$$v_{ex} = I_{sp} \cdot g_0 = 253 \times 9.8066 = 2481 \text{ m/s}$$

m_0 = chaser wet mass at the start of the phase [kg]

(decreases through the mission as prop is burned)

$I_{sp} = 253 \text{ s}$ (MS monoprop, confirmed)

KOS1 derivation (per-target, uplinked by GO at cut-off)

$g_0 = 9.8066 \text{ m/s}^2$ (standard gravity)

$$r_{KOS1} = \frac{s_{panel}}{2} \times \kappa_{margin}$$

s_{panel} = solar-panel tip-to-tip span, confirmed per target from GO uplink

$\kappa_{margin} = 1.20$ (+20% safety margin)

COMSATBW-1 (BSS-702, confirmed): $17.2/2 \times 1.20 = 10.32 \text{ m}$

KOS2 derivation (fixed geometry)

$$r_{KOS2} = L_{arm} + \frac{L_{tug}}{2}$$

$L_{arm} = 5.0 \text{ m}$ (arm length TBD MIRON)

$L_{tug} = 0.8 \text{ m}$ (tug body length, confirmed)

Current value: $5.0 + 0.4 = 5.40 \text{ m}$ (auto-updates when MIRON provides arm length)